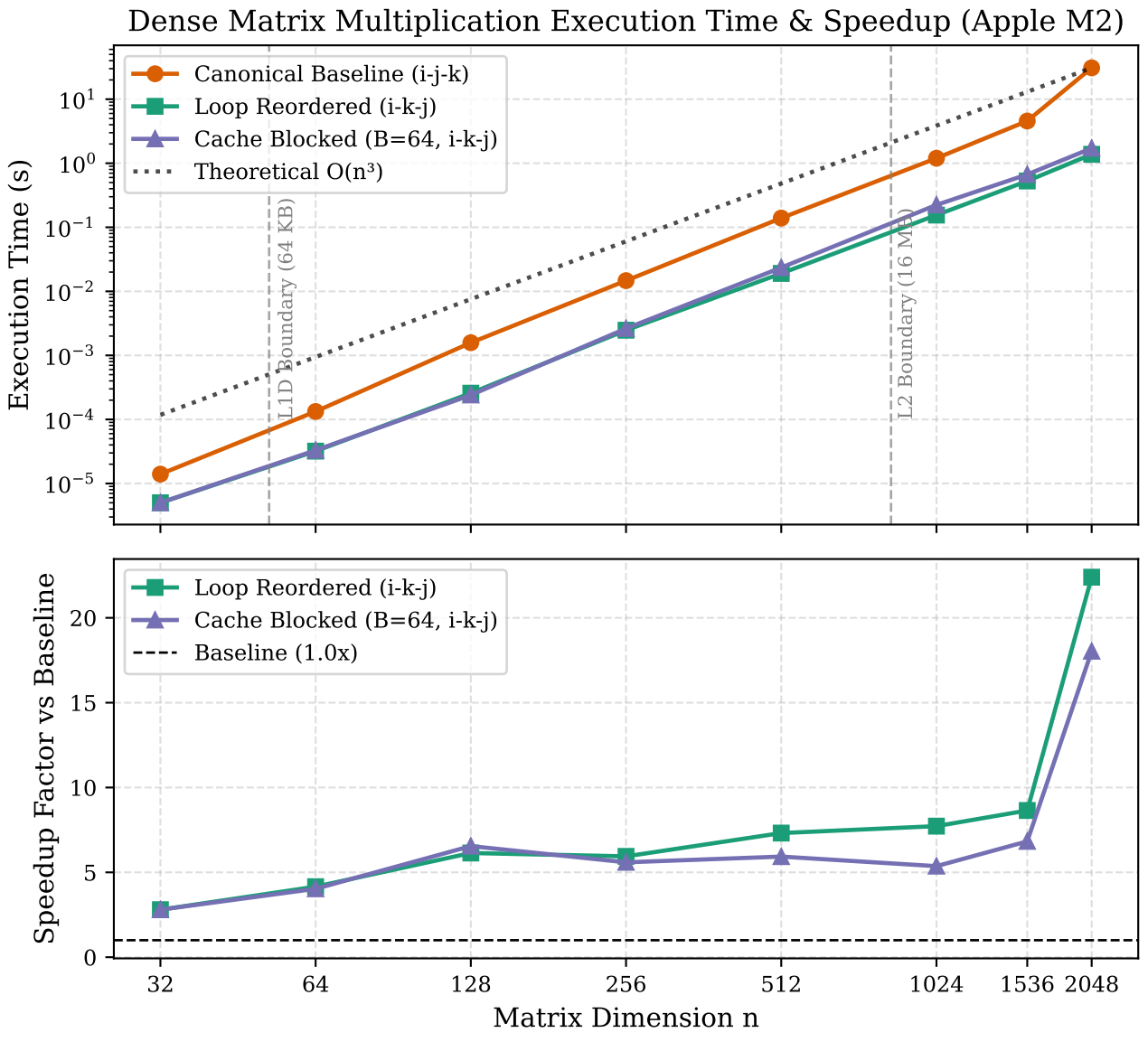
Dense Matrix Multiplication Execution Time & Speedup (Apple M2)
Execution Time (s)
Canonical Baseline (i-j-k)
Loop Reordered (i-k-j)
Cache Blocked (B=64, i-k-j)
Theoretical O(n³)
L1D Boundary (64 KB)
L2 Boundary (16 MB)
Speedup Factor vs Baseline
Loop Reordered (i-k-j)
Cache Blocked (B=64, i-k-j)
Baseline (1.0x)
Matrix Dimension n
32
64
128
256
512
1024
1536
2048